

Ch 4: Carbon and Its Compounds

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Why is carbon able to form a very large number of compounds? State two special properties of carbon responsible for this. (PYQ 2022)

Q2. What is catenation? Give one example of a compound formed due to catenation. Which element shows catenation to the maximum extent?

Q3. What is tetravalency of carbon? How does it help carbon form so many compounds? (PYQ 2020)

Q4. What is a covalent bond? What are the two types of covalent bonds? Give one example of each.

Q5. What are allotropes of carbon? Name three allotropes of carbon and state one property of each. (PYQ 2019)

Q6. Why does diamond have a very high melting point while graphite is a good conductor of electricity? (PYQ 2022)

Q7. What is a homologous series? State any two characteristics of a homologous series. (PYQ 2019)

Q8. Write the molecular formula of the first two members of the homologous series having functional group: (i) -OH (ii) -COOH. (PYQ 2020)

Q9. What are saturated hydrocarbons? What are unsaturated hydrocarbons? Give one example of each. (PYQ 2019)

Q10. What is the general formula for alkanes, alkenes, and alkynes? Give one example of each series. (PYQ 2022)

Q11. Name the functional group present in each of the following compounds: (i) CH_3COCH_3 (ii) CH_3CHO (iii) CH_3COOH (iv) CH_3OH . (PYQ 2020)

Q12. Write the IUPAC names of the following compounds: (i) $\text{CH}_3\text{CH}_2\text{OH}$ (ii) CH_3COOH (iii) CH_3CHO (iv) HCOOH . (PYQ 2019)

Q13. Write the structural formulae of: (i) Ethane (ii) Propene (iii) Ethyne (iv) Butanone. (PYQ 2022)

Q14. What is isomerism? Give one example of two compounds that are isomers of each other. Write their structural formulae. (PYQ 2019)

Q15. What is combustion of carbon compounds? Write the balanced equation for complete combustion of ethane. (PYQ 2020)

Q16. What is oxidation in carbon compounds? Write the balanced equation for the oxidation of ethanol using acidified potassium dichromate. (PYQ 2019)

Q17. What is an addition reaction? Give one example of an addition reaction of ethene with hydrogen. Write the balanced equation. (PYQ 2022)

Q18. What is a substitution reaction? Give one example of a substitution reaction of methane with chlorine. Write the balanced equation. (PYQ 2020)

Q19. What is hydrogenation? What is the industrial importance of hydrogenation? Write the balanced equation for hydrogenation of ethene. (PYQ 2019)

Q20. What is ethanol? Write its molecular formula. State two physical properties of ethanol. (PYQ 2020)

Q21. What happens when ethanol is heated with concentrated H_2SO_4 at 170 degrees Celsius? Write the balanced chemical equation. (PYQ 2019)

Q22. What happens when ethanol reacts with sodium metal? Write the balanced chemical equation and name the gas evolved. (PYQ 2022)

Q23. What is denatured alcohol? Why is industrial ethanol denatured? What substances are added to denature it? (PYQ 2020)

Q24. What is ethanoic acid? Write its molecular formula. Why is it called glacial acetic acid? (PYQ 2019)

Q25. What happens when ethanoic acid reacts with sodium carbonate? Write the balanced equation and name the gas evolved. (PYQ 2022)

Q26. What is esterification? Write the balanced chemical equation for the reaction between ethanol and ethanoic acid. Name the product formed. (PYQ 2019)

Q27. What is saponification? Write the balanced chemical equation for saponification of an ester. (PYQ 2020)

Q28. What are soaps? How are soaps made? Name the type of chemical reaction involved in soap making. (PYQ 2022)

Q29. What are detergents? How are they different from soaps in terms of their effectiveness in hard water? (PYQ 2019)

Q30. Why do soaps form scum in hard water? Why do detergents not form scum in hard water? (PYQ 2020)

Q31. What is micelle formation? Draw a labelled diagram of a micelle and explain how it helps in cleansing. (PYQ 2019)

Q32. Select the saturated hydrocarbons from the following: C_3H_6 , C_5H_{10} , C_4H_{10} , C_6H_{14} , C_2H_4 . Justify your selection. (PYQ 2022)

SECTION B - 3 MARKS

Q1. Define a homologous series. State three characteristics of a homologous series. Write the first four members of the alkane homologous series with their molecular formulae. (PYQ 2019)

Q2. Complete the following table about functional groups and the corresponding homologous series: (PYQ 2020)

Functional Group	Name of Functional Group	General Formula	IUPAC Suffix/Prefix	Example (name + formula)
-OH				
-CHO				
>C=O (ketone)				
-COOH				
-Cl / -Br				

Q3. Write the IUPAC names of the following compounds: (PYQ 2022)

Structural Formula	IUPAC Name	Functional Group Present
CH ₃ -CH ₂ -OH		
CH ₃ -CO-CH ₃		
CH ₃ -CH ₂ -CHO		
CH ₃ -CH ₂ -COOH		
CH ₂ =CH ₂		

Q4. What are the four types of chemical reactions shown by carbon compounds? Write one balanced chemical equation and give one example for each type: (PYQ 2019)

Type of Reaction	Balanced Chemical Equation	Example Compound
Combustion		
Oxidation		
Addition		

Type of Reaction	Balanced Chemical Equation	Example Compound
Substitution		

Q5. Write the balanced chemical equations for the following reactions of ethanoic acid: (PYQ 2020)

Substance Reacting with Ethanoic Acid	Balanced Chemical Equation	Product(s) Formed
Sodium (Na) metal		
Sodium carbonate (Na ₂ CO ₃)		
Sodium hydroxide (NaOH)		
Ethanol (C ₂ H ₅ OH)		

Q6. Write the balanced chemical equations for the following reactions of ethanol: (PYQ 2019)

Reaction of Ethanol	Balanced Chemical Equation	Product(s) Formed
Burning (combustion) in air		
With sodium metal		
Oxidation (with acidified K ₂ Cr ₂ O ₇)		
Dehydration (conc. H ₂ SO ₄ , 170 C)		

Q7. Write the structural formula and IUPAC name for the following: (PYQ 2022)

Description	Structural Formula	IUPAC Name
An alcohol with 3 carbon atoms		
A carboxylic acid with 2 carbon atoms		
An aldehyde with 4 carbon atoms		
A ketone with 4 carbon atoms		
An alkene with 3 carbon atoms		

Q8. Write the next homologue of each of the following and name it: (PYQ 2020)

Compound	Molecular Formula	Next Homologue Formula	Name of Next Homologue
Methane (CH ₄)	CH ₄		
Ethene (C ₂ H ₄)	C ₂ H ₄		
Ethyne (C ₂ H ₂)	C ₂ H ₂		
Methanol (CH ₃ OH)	CH ₃ OH		
Ethanoic acid (CH ₃ COOH)	CH ₃ COOH		

Q9. Compare soaps and detergents on the following basis: (PYQ 2022)

Basis of Comparison	Soaps	Detergents
Prepared from		
Nature of salt (sodium / ammonium)		
Effectiveness in hard water		
Formation of scum		
Environmental impact (biodegradable?)		

Q10. A saturated organic compound A belongs to the homologous series of alkanes. Its molecular formula is C₄H₁₀. (PYQ 2023)

Question	Answer
Write the name of compound A	
Write two possible structural formulae of A (isomers)	
Name both the isomers	
What is the relationship between these two structures?	

Q11. Why is carbon able to form such a large variety of compounds? Explain catenation and tetravalency with one example each. How are these two properties different from each other? (PYQ 2022)

Q12. Explain the structure and properties of diamond and graphite. Why is diamond used as a cutting tool while graphite is used as a lubricant? (PYQ 2019)

Q13. Write the balanced equations for the following. Also state the type of reaction in each case: (PYQ 2024)

Reaction	Balanced Equation	Type of Reaction
$\text{CH}_4 + \text{Cl}_2$ (in sunlight) $\rightarrow$		
$\text{CH}_2=\text{CH}_2 + \text{H}_2$ (Ni catalyst) $\rightarrow$		
$\text{C}_2\text{H}_5\text{OH} + \text{O}_2$ (excess) $\rightarrow$		
$\text{C}_2\text{H}_5\text{OH} + \text{CH}_3\text{COOH}$ (H^+ catalyst) $\rightarrow$		
$\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \rightarrow$		

Q14. Explain how soaps and detergents clean oily dirt from clothes. Draw the structure of a soap molecule and label the hydrophilic and hydrophobic parts. What is a micelle? (PYQ 2020)

Q15. Write the structural formulas and names of the first three members of the following homologous series: (PYQ 2019)

Homologous Series	Member 1 (formula + name)	Member 2 (formula + name)	Member 3 (formula + name)
Alcohols (-OH)			
Aldehydes (-CHO)			
Carboxylic Acids (-COOH)			

Q16. What is esterification? What is saponification? Write balanced equations for both reactions. What is the catalyst used in esterification and what is it called? (PYQ 2022)

Q17. What is denatured alcohol? Why is it used? What are the ill effects of drinking ethanol? What happens when ethanol is consumed in large quantities? (PYQ 2020)

Q18. Name the following compounds and write their IUPAC names: (PYQ 2019)

Molecular Formula	Class of Compound	IUPAC Name	Common Name (if any)
CH ₃ OH			
C ₂ H ₅ OH			
CH ₃ COOH			
HCOOH			
CH ₃ CHO			

Q19. Distinguish between saturated and unsaturated hydrocarbons on the basis of: (i) type of bonds (ii) general formula (iii) burning (iv) addition reaction (v) example. (PYQ 2022)

Basis	Saturated Hydrocarbons	Unsaturated Hydrocarbons
Type of bond between C atoms		
General formula (for acyclic)		
Nature of flame (burning)		
Addition reaction		
Example		

Q20. Write the molecular formulae and draw the structural formulae of the following: (PYQ 2020)

Name of Compound	Molecular Formula	Structural Formula (expanded)
Ethanoic acid		
Butanone		
Propanal		
But-2-ene		
Propan-1-ol		

Q21. Explain what happens when: (i) ethanol is heated with concentrated H₂SO₄ at 170 C (ii) ethanol is treated with acidified potassium dichromate (iii) ethanol is mixed with sodium metal. Write equations for each. (PYQ 2019)

Q22. What is the cleansing action of soaps? Explain with the help of a labelled diagram of a micelle. Why do soaps not work well in hard water? (PYQ 2022)

Q23. What is glacial acetic acid? Write three properties of ethanoic acid (acetic acid). Write equations for its reactions with: (i) Na_2CO_3 (ii) NaOH . (PYQ 2020)

Q24. Write the full structural formulae and IUPAC names of the two isomers of butane (C_4H_{10}). State the type of isomerism shown. How do they differ in boiling points? (PYQ 2019)

Q25. Write balanced chemical equations for the complete combustion of the following compounds in excess air: (PYQ 2022)

Compound	Balanced Equation for Combustion	Products Formed
Methane (CH_4)		
Ethanol ($\text{C}_2\text{H}_5\text{OH}$)		
Ethanoic acid (CH_3COOH)		
Propane (C_3H_8)		
Ethene (C_2H_4)		

Q26. Explain why: (i) carbon forms compounds mainly by covalent bonding (ii) covalent compounds have low melting and boiling points (iii) they do not conduct electricity in general. (PYQ 2023)

Q27. What is the difference between a soap and a detergent in terms of chemical composition? Why are biodegradable detergents preferred over non-biodegradable detergents? (PYQ 2020)

Q28. Name the following and write the IUPAC name of each: (PYQ 2024)

Description	IUPAC Name	Molecular Formula
An alkane with 5 carbon atoms		
An alkene with 4 carbon atoms		
An alkyne with 3 carbon atoms		
A primary alcohol with 4 carbon atoms		
A carboxylic acid with 3 carbon atoms		

Q29. What are the uses of ethanol in everyday life? State three uses. What are the harmful effects of drinking alcohol? Write the reaction of ethanol with sodium. (PYQ 2019)

Q30. Explain why: (i) soaps are not suitable for washing in hard water (ii) detergents are better than soaps for washing in hard water (iii) oils should be chosen that are rich in unsaturated fatty acids for cooking. (PYQ 2020)

SECTION C - 5 MARKS

Q1. Explain why carbon forms such a large number of compounds. Describe the special properties of carbon (catenation and tetravalency) with examples. Also name the allotropes of carbon and explain the structure and properties of diamond and graphite. (PYQ 2022)

Q2. What is a homologous series? Write three characteristics of a homologous series. Write the first four members and general formulae for the following series: (PYQ 2019)

Homologous Series	General Formula	First 4 Members with Formulae and Names
Alkanes		
Alkenes		
Alkynes		
Alcohols		
Carboxylic Acids		

Q3. Write the balanced chemical equations for all four types of reactions shown by carbon compounds. Give one example of each. Also state the conditions (temperature, catalyst, etc.) required: (PYQ 2020)

Type of Reaction	Balanced Chemical Equation	Condition Required	Product(s) Formed
Combustion			
Oxidation			
Addition reaction			
Substitution reaction			

Q4. Write the balanced chemical equations for the following reactions of ethanoic acid and ethanol. State one observation for each: (PYQ 2020)

Reaction	Balanced Chemical Equation	Observation
Ethanoic acid + Sodium metal		
Ethanoic acid + Na ₂ CO ₃		
Ethanoic acid + NaOH		

Reaction	Balanced Chemical Equation	Observation
Ethanoic acid + Ethanol (H ⁺ catalyst)		
Ethanol + Sodium metal		
Ethanol + Acidified K ₂ Cr ₂ O ₇		
Ethanol + Conc. H ₂ SO ₄ at 170 °C		

Q5. Complete the following table about important carbon compounds: their names, formulae, structural formulae, and uses: (PYQ 2022)

Common Name	IUPAC Name	Molecular Formula	Structural Formula (expanded)	One Important Use
Methane				
Ethylene				
Acetylene				
Ethanol				
Acetic acid				

Q6. Explain the cleansing action of soaps. Draw a labelled diagram of a micelle. Explain why soaps do not work in hard water. Compare soaps and detergents across five properties. (PYQ 2019)

Property	Soaps	Detergents
Prepared from		
Type of salt		
Works in hard water?		
Forms scum?		
Biodegradable?		

Q7. A student is given an organic compound A with molecular formula C₂H₆O. (PYQ 2023)

Question	Answer
Write the IUPAC name of A	

Question	Answer
Write the structural formula of A	
What happens when A reacts with sodium? Write equation.	
What happens when A is oxidised with acidified $K_2Cr_2O_7$? Write equation.	
What happens when A is heated with conc. H_2SO_4 ? Write equation.	

Q8. Write the IUPAC names and structural formulae for the following compounds. Also identify the functional group and class of compound in each: (PYQ 2024)

Molecular Formula	IUPAC Name	Structural Formula	Functional Group	Class of Compound
C_3H_7OH				
C_3H_6O (aldehyde)				
C_4H_8O (ketone)				
$C_3H_6O_2$				
C_4H_6				

Q9. Explain the structure of diamond and graphite. Write three differences between them. Why is graphite a good conductor of electricity while diamond is not? (PYQ 2022)

Basis	Diamond	Graphite
Structure / Arrangement of carbon atoms		
Type of bonding		
Free electrons		
Hardness		
Electrical conductivity		
Uses		

Q10. What is isomerism? Write the structural formulae and IUPAC names of all possible isomers of C₄H₁₀ and C₄H₈ (alkene). State the type of isomerism. (PYQ 2019)

Q11. Write the balanced chemical equations for the following reactions. State the type of reaction in each case and identify the functional group that undergoes the reaction: (PYQ 2020)

Reaction	Balanced Equation	Type	Functional Group Involved
Ethene + H ₂ (Ni catalyst)			
Ethyne + Br ₂ (in CCl ₄)			
Methane + Cl ₂ (sunlight)			
Ethanol + Na			
Ethanoic acid + NaOH			

Q12. Describe the properties of ethanoic acid. Write balanced equations for its reactions with: (i) sodium (ii) sodium carbonate (iii) sodium hydroxide (iv) ethanol. Also state what happens when ethanoic acid is cooled below 16.6 degrees Celsius. (PYQ 2019)

Q13. Case Study: A student is studying a compound X. It is a liquid at room temperature with a characteristic smell. When heated with conc. H₂SO₄ at 170 °C it forms compound Y. When oxidised with acidified K₂Cr₂O₇ it forms compound Z. Compound X reacts with Z in presence of H₂SO₄ to form a sweet-smelling compound W. (PYQ 2024 case-based)

Question	Answer
Identify compound X (with molecular formula)	
Identify compound Y and write its molecular formula	
Identify compound Z and write its molecular formula	
Identify compound W and write the balanced equation for its formation	
Name the type of reaction by which X forms Z	

Q14. Write the structural formulae, molecular formulae, and IUPAC names for the first four members of each of the following homologous series: (PYQ 2022)

Series	Member 1	Member 2	Member 3	Member 4
Alkanes (C_nH_{2n+2})				
Alcohols ($C_nH_{2n+1}OH$)				
Carboxylic acids				

Q15. What are soaps and detergents? Explain how soaps are made (saponification). Draw a labelled diagram of a soap molecule showing its hydrophilic and hydrophobic ends. Explain the micelle formation and how it helps in cleansing action. Why are detergents preferred over soaps in hard water? (PYQ 2020)